The Ultimate Python 300 MCQ Workbook

Extended Edition: Focus on Functions, Default Arguments, and Core Logic

1. What is the output of the following code?

```
def serf(x, y=6):
    return(x+y)

print(serf(3))
```

a. 6
b. 3
c. Error
d. 9

Correct Answer: 9
Reasoning: When serf(3) is called, 'x' is assigned 3. Since no second argument is provided, 'y' falls back to its default value of 6. 3 + 6 = 9.

---

2. What is the output?

```
power = lambda x, y: x ** y
print(power(3, 3))
```

a. 9
b. 48
c. Error
d. 27

Correct Answer: 27
Reasoning: The lambda function returns x raised to the power of y. 3 ** 3 = 27.

---

3. What is the output of print(3 + 2 * 2)?

a. 6
b. Error
c. 7
d. 10

Correct Answer: 7
Reasoning: Multiplication has higher precedence than addition, so it evaluates as a + (b * c).

---

4. What is the result of print('b' * 6)?

a. bbbbbb
b. 6
c. Error
d. b

Correct Answer: bbbbbb
Reasoning: Multiplying a string by an integer repeats the string that many times.

---

5. What is the output?

```
def my_func(**kwargs):
    return len(kwargs)

print(my_func(a=1, b=2, c=3))
```

a. 1
b. 0
c. Error
d. 3

Correct Answer: 3
Reasoning: **kwargs collects keyword arguments into a dictionary. 3 pairs were passed.

---

6. What is the output?

```
d = {'B': 2}
print(d.get('X', 8))
```

a. None
b. 2
c. Error
d. 8

Correct Answer: 8
Reasoning: The get() method returns the default value (8) because the key 'X' is not in the dictionary.

---

7. What is the output?

```
def multiply(a, b=4):
    return a * b

print(multiply(4, 10))
```

a. Error
b. 16
c. 40
d. 10

Correct Answer: 40
Reasoning: The default value 4 is overridden by 10. 4 * 10 = 40.

---

8. What is the result of print('a' * 6)?

a. Error
b. a
c. 6
d. aaaaaa

Correct Answer: aaaaaa
Reasoning: Multiplying a string by an integer repeats the string that many times.

---

9. What is the output of print({1, 2} | {2, 3})?

a. {2}
b. {1, 2, 3}
c. Error
d. {1, 3}

Correct Answer: {1, 2, 3}
Reasoning: The | operator performs a set union, combining all unique elements.

---

10. What is printed by the last line?

```
def append_to(num, target=[]):
    target.append(num)
    return target

append_to(99)
print(append_to(1))
```

a. [1]
b. Error
c. [99]
d. [99, 1]

Correct Answer: [99, 1]
Reasoning: Default list arguments are created once at definition. The list persists across function calls.

---

11. What is the output?

```
def my_func(**kwargs):
    return len(kwargs)

print(my_func(a=1, b=2, c=3))
```

a. 0
b. Error
c. 3
d. 1

Correct Answer: 3
Reasoning: **kwargs collects keyword arguments into a dictionary. 3 pairs were passed.

---

12. What is the output of print({1, 2} | {2, 3})?

a. Error
b. {1, 3}
c. {2}
d. {1, 2, 3}

Correct Answer: {1, 2, 3}
Reasoning: The | operator performs a set union, combining all unique elements.

---

13. What is the output?

```
nums = [x * 4 for x in range(3)]
print(nums)
```

a. Error
b. [4, 8, 12]
c. [0, 4, 8]
d. [0, 4, 8, 12]

Correct Answer: [0, 4, 8]
Reasoning: range(3) yields 0, 1, 2. Multiplying each by 4 results in [0, 4, 8].

---

14. What is the output?

```
d = {'A': 3}
print(d.get('Z', 6))
```

a. Error
b. 6
c. None
d. 3

Correct Answer: 6
Reasoning: The get() method returns the default value (6) because the key 'Z' is not in the dictionary.

---

15. What does this code print?

```
def count_items(*args):
    return len(args)

print(count_items(7, 5, 9, 4, 6))
```

a. 4
b. 6
c. Error
d. 5

Correct Answer: 5
Reasoning: *args collects positional arguments into a tuple. There are 5 arguments passed.

---

16. What is the output?

```
def my_func(**kwargs):
    return len(kwargs)

print(my_func(a=1, b=2, c=3))
```

a. 3
b. Error
c. 0
d. 1

Correct Answer: 3
Reasoning: **kwargs collects keyword arguments into a dictionary. 3 pairs were passed.

---

17. What is the output?

```
def multiply(a, b=4):
    return a * b

print(multiply(3, 7))
```

a. 21
b. 7
c. 12
d. Error

Correct Answer: 21
Reasoning: The default value 4 is overridden by 7. 3 * 7 = 21.

---

18. What does this code print?

```
def count_items(*args):
    return len(args)

print(count_items(3, 4, 9, 9))
```

a. 4
b. 3
c. Error
d. 5

Correct Answer: 4
Reasoning: *args collects positional arguments into a tuple. There are 4 arguments passed.

---

19. What is the output of print('banana'.find('na'))?

a. 0
b. 2
c. 4
d. -1

Correct Answer: 2
Reasoning: find() returns the lowest index of the substring. 'na' starts at index 2.

---

20. What does this code print?

```
def count_items(*args):
    return len(args)

print(count_items(8, 2, 1, 2, 3, 3))
```

a. 5
b. 6
c. 7
d. Error

Correct Answer: 6
Reasoning: *args collects positional arguments into a tuple. There are 6 arguments passed.

---

21. What does this evaluate to?

```
print(False and False)
```

a. True
b. False
c. Error
d. None

Correct Answer: False
Reasoning: The 'and' operator requires both operands to be True. False and False is False.

---

22. What is the output of print(type('123'))?

a.
b.
c. Error
d.

Correct Answer:
Reasoning: The value '123' is recognized by Python as .

---

23. What is the output?

```
def multiply(a, b=5):
    return a * b

print(multiply(4, 7))
```

a. 7
b. 20
c. 28
d. Error

Correct Answer: 28
Reasoning: The default value 5 is overridden by 7. 4 * 7 = 28.

---

24. What is the result of print('b' * 3)?

a. b
b. bbb
c. Error
d. 3

Correct Answer: bbb
Reasoning: Multiplying a string by an integer repeats the string that many times.

---

25. What does this code print?

```
def count_items(*args):
    return len(args)

print(count_items(3, 9, 9, 1))
```

a. Error
b. 4
c. 3
d. 5

Correct Answer: 4
Reasoning: *args collects positional arguments into a tuple. There are 4 arguments passed.

---

26. What is the output?

```
def multiply(a, b=4):
    return a * b

print(multiply(4, 7))
```

a. 16
b. 7
c. Error
d. 28

Correct Answer: 28
Reasoning: The default value 4 is overridden by 7. 4 * 7 = 28.

---

27. What is the output?

```
def multiply(a, b=2):
    return a * b

print(multiply(5, 10))
```

a. 50
b. 9
c. Error
d. 10

Correct Answer: 50
Reasoning: The default value 2 is overridden by 10. 5 * 10 = 50.

---

28. What is the result of print('b' * 5)?

a. b
b. 5
c. bbbbb
d. Error

Correct Answer: bbbbb
Reasoning: Multiplying a string by an integer repeats the string that many times.

---

29. What is printed by the last line?

```
def append_to(num, target=[]):
    target.append(num)
    return target

append_to(99)
print(append_to(3))
```

a. [99, 3]
b. [3]
c. [99]
d. Error

Correct Answer: [99, 3]
Reasoning: Default list arguments are created once at definition. The list persists across function calls.

---

30. What is the output?

```
nums = [x * 4 for x in range(3)]
print(nums)
```

a. [4, 8, 12]
b. [0, 4, 8, 12]
c. [0, 4, 8]
d. Error

Correct Answer: [0, 4, 8]
Reasoning: range(3) yields 0, 1, 2. Multiplying each by 4 results in [0, 4, 8].

---

31. What is the output?

```
power = lambda x, y: x ** y
print(power(2, 4))
```

a. 2
b. 16
c. Error
d. 8

Correct Answer: 16
Reasoning: The lambda function returns x raised to the power of y. 2 ** 4 = 16.

---

32. What is the output?

```
def calc(a, b=2):
    return a + b

print(calc(3))
```

a. 7
b. 2
c. 5
d. 3

Correct Answer: 5
Reasoning: Parameter 'b' defaults to 2. 3 + 2 = 5.

---

33. What is the output of print({1, 2} | {2, 3})?

a. {2}
b. Error
c. {1, 2, 3}
d. {1, 3}

Correct Answer: {1, 2, 3}
Reasoning: The | operator performs a set union, combining all unique elements.

---

34. If data = [10, 11, 12, 13, 14, 15, 16], what is the output of print(data[2:4])?

a. Error
b. [12, 13]
c. [13]
d. [12, 13, 14]

Correct Answer: [12, 13]
Reasoning: Slicing [start:end] includes the start index (2) but excludes the end index (4).

---

35. What is printed by the last line?

```
def append_to(num, target=[]):
    target.append(num)
    return target

append_to(99)
print(append_to(5))
```

a. [99]
b. [5]
c. [99, 5]
d. Error

Correct Answer: [99, 5]
Reasoning: Default list arguments are created once at definition. The list persists across function calls.

---

36. What is the output?

```
def my_func(**kwargs):
    return len(kwargs)

print(my_func(a=1, b=2, c=3))
```

a. 0
b. 1
c. 3
d. Error

Correct Answer: 3
Reasoning: **kwargs collects keyword arguments into a dictionary. 3 pairs were passed.

---

37. What is the output?

```
def multiply(a, b=4):
    return a * b

print(multiply(5, 9))
```

a. 9
b. 45
c. 20
d. Error

Correct Answer: 45
Reasoning: The default value 4 is overridden by 9. 5 * 9 = 45.

---

38. What is the output of print(type('123'))?

a. Error

b.

c.

Correct Answer:
Reasoning: The value '123' is recognized by Python as .

---

39. What is the output?

```
def my_func(**kwargs):
    return len(kwargs)

print(my_func(a=1, b=2, c=3))
```

a. 0

b. 1

c. Error

d. 3

Correct Answer: 3
Reasoning: **kwargs collects keyword arguments into a dictionary. 3 pairs were passed.

---

40. What is the output?

```
power = lambda x, y: x ** y
print(power(5, 3))
```

a. 243

b. 15

c. 125

d. Error

Correct Answer: 125
Reasoning: The lambda function returns x raised to the power of y. 5 ** 3 = 125.

---

41. What is the output?

```
nums = [x * 2 for x in range(3)]
print(nums)
```

a. [0, 2, 4, 6]

b. Error

c. [2, 4, 6]

d. [0, 2, 4]

Correct Answer: [0, 2, 4]
Reasoning: range(3) yields 0, 1, 2. Multiplying each by 2 results in [0, 2, 4].

---

42. What is the output?

```
def calc(a, b=2):
    return a + b

print(calc(10))
```

a. 10
b. 2
c. 14
d. 12

Correct Answer: 12
Reasoning: Parameter 'b' defaults to 2. 10 + 2 = 12.

---

43. What does this evaluate to?

```
print(False and False)
```

a. None
b. False
c. Error
d. True

Correct Answer: False
Reasoning: The 'and' operator requires both operands to be True. False and False is False.

---

44. What does this code print?

```
def count_items(*args):
    return len(args)

print(count_items(1, 7, 5, 8, 5))
```

a. 5
b. 4
c. 6
d. Error

Correct Answer: 5
Reasoning: *args collects positional arguments into a tuple. There are 5 arguments passed.

---

45. What does this code print?

```
def count_items(*args):
    return len(args)

print(count_items(5, 4, 1))
```

a. Error
b. 2
c. 4
d. 3

Correct Answer: 3
Reasoning: *args collects positional arguments into a tuple. There are 3 arguments passed.

---

46. What is the output?

```
def calc(a, b=9):
    return a + b

print(calc(2))
```

a. 13
b. 9
c. 11
d. 2

Correct Answer: 11
Reasoning: Parameter 'b' defaults to 9. 2 + 9 = 11.

---

47. What is the output of print({1, 2} | {2, 3})?

a. Error
b. {1, 3}
c. {1, 2, 3}
d. {2}

Correct Answer: {1, 2, 3}
Reasoning: The | operator performs a set union, combining all unique elements.

---

48. What is the output of print(type(45.0))?

a.
b.
c. Error

Correct Answer:
Reasoning: The value 45.0 is recognized by Python as .

---

49. What is the output of print(4 + 2 * 4)?

a. Error
b. 10
c. 24
d. 12

Correct Answer: 12
Reasoning: Multiplication has higher precedence than addition, so it evaluates as a + (b * c).

---

50. What is the output of print('banana'.find('na'))?

a. -1
b. 4
c. 2
d. 0

Correct Answer: 2
Reasoning: find() returns the lowest index of the substring. 'na' starts at index 2.

---

51. What is the output?

```
power = lambda x, y: x ** y
print(power(4, 5))
```

a. 1024
b. Error
c. 625
d. 20

Correct Answer: 1024
Reasoning: The lambda function returns x raised to the power of y. 4 ** 5 = 1024.

---

52. What is the output?

```
nums = [x * 3 for x in range(3)]
print(nums)
```

a. [0, 3, 6, 9]
b. [3, 6, 9]
c. Error
d. [0, 3, 6]

Correct Answer: [0, 3, 6]
Reasoning: range(3) yields 0, 1, 2. Multiplying each by 3 results in [0, 3, 6].

---

53. What is the output of print(4 + 2 * 2)?

a. 8
b. 35
c. Error
d. 12

Correct Answer: 8
Reasoning: Multiplication has higher precedence than addition, so it evaluates as a + (b * c).

---

54. What does this code print?

```
def count_items(*args):
    return len(args)

print(count_items(2, 4, 6, 5))
```

a. 4
b. 5
c. Error
d. 3

Correct Answer: 4
Reasoning: *args collects positional arguments into a tuple. There are 4 arguments passed.

---

55. What is the output of print(4 + 4 * 2)?

a. 16
b. 12
c. Error
d. 10

Correct Answer: 12
Reasoning: Multiplication has higher precedence than addition, so it evaluates as a + (b * c).

---

56. What is the output of print('banana'.find('na'))?

a. 0
b. 2
c. -1
d. 4

Correct Answer: 2
Reasoning: find() returns the lowest index of the substring. 'na' starts at index 2.

---

57. What is the output of print('banana'.find('na'))?

a. 4
b. -1
c. 0
d. 2

Correct Answer: 2
Reasoning: find() returns the lowest index of the substring. 'na' starts at index 2.

---

58. If data = [10, 11, 12, 13, 14, 15, 16], what is the output of print(data[1:5])?

a. [11, 12, 13, 14]
b. Error
c. [12, 13, 14]
d. [11, 12, 13, 14, 15]

Correct Answer: [11, 12, 13, 14]
Reasoning: Slicing [start:end] includes the start index (1) but excludes the end index (5).

---

59. What is the output?

```
power = lambda x, y: x ** y
print(power(4, 5))
```

a. 20
b. Error
c. 1024
d. 625

Correct Answer: 1024
Reasoning: The lambda function returns x raised to the power of y. 4 ** 5 = 1024.

---

60. What is the output of print(type((1, 2)))?

a. <class 'tuple'>
b. Error
c. <class 'int'>

Correct Answer:
Reasoning: The value (1, 2) is recognized by Python as .

---

61. What is the output?

```
d = {'A': 3}
print(d.get('Z', 7))
```

a. 7
b. 3
c. None
d. Error

Correct Answer: 7
Reasoning: The get() method returns the default value (7) because the key 'Z' is not in the dictionary.

---

62. What is the result of print('a' * 3)?

a. a
b. Error
c. 3
d. aaa

Correct Answer: aaa
Reasoning: Multiplying a string by an integer repeats the string that many times.

---

63. What is the result of print('a' * 5)?

a. 5
b. aaaaa
c. Error
d. a

Correct Answer: aaaaa
Reasoning: Multiplying a string by an integer repeats the string that many times.

---

64. What is the output?

```
def calc(a, b=7):
    return a + b

print(calc(6))
```

a. 7
b. 15
c. 6
d. 13

Correct Answer: 13
Reasoning: Parameter 'b' defaults to 7. 6 + 7 = 13.

---

65. What is the output of print(type(45.0))?

a. Error
b.
c.

Correct Answer:
Reasoning: The value 45.0 is recognized by Python as .

---

66. What is the output of print(4 + 2 * 2)?

a. Error
b. 8
c. 11
d. 12

Correct Answer: 8
Reasoning: Multiplication has higher precedence than addition, so it evaluates as a + (b * c).

---

67. What does this code print?

```
def count_items(*args):
    return len(args)

print(count_items(7, 4, 5, 3))
```

a. 5
b. Error
c. 3
d. 4

Correct Answer: 4
Reasoning: *args collects positional arguments into a tuple. There are 4 arguments passed.

---

68. What is the output of print({1, 2} | {2, 3})?

a. {2}
b. {1, 3}
c. {1, 2, 3}
d. Error

Correct Answer: {1, 2, 3}
Reasoning: The | operator performs a set union, combining all unique elements.

---

69. What is the output of print({1, 2} | {2, 3})?

a. {1, 2, 3}
b. {1, 3}
c. {2}
d. Error

Correct Answer: {1, 2, 3}
Reasoning: The | operator performs a set union, combining all unique elements.

---

70. What is the output of print({1, 2} | {2, 3})?

a. Error
b. {1, 3}
c. {1, 2, 3}
d. {2}

Correct Answer: {1, 2, 3}
Reasoning: The | operator performs a set union, combining all unique elements.

---

71. What is the output of print(type(45.0))?

b.
c. Error

Correct Answer:
Reasoning: The value 45.0 is recognized by Python as .

---

72. What is the output of print({1, 2} | {2, 3})?

a. Error
b. {1, 3}
c. {2}
d. {1, 2, 3}

Correct Answer: {1, 2, 3}
Reasoning: The | operator performs a set union, combining all unique elements.

---

73. What is the output of print(type((1, 2)))?

b. Error
c.
d.

Correct Answer:
Reasoning: The value (1, 2) is recognized by Python as .

---

74. What is the output of print('banana'.find('na'))?

a. -1
b. 4
c. 2
d. 0

Correct Answer: 2
Reasoning: find() returns the lowest index of the substring. 'na' starts at index 2.

---

75. What is the output?

```
def calc(a, b=8):
    return a + b

print(calc(3))
```

a. 11
b. 3
c. 13
d. 8

Correct Answer: 11
Reasoning: Parameter 'b' defaults to 8. 3 + 8 = 11.

---

76. What does this evaluate to?

```
print(True and False)
```

a. None
b. False
c. Error
d. True

Correct Answer: False
Reasoning: The 'and' operator requires both operands to be True. True and False is False.

---

77. What is printed by the last line?

```
def append_to(num, target=[]):
    target.append(num)
    return target

append_to(99)
print(append_to(4))
```

a. [4]
b. Error
c. [99]
d. [99, 4]

Correct Answer: [99, 4]
Reasoning: Default list arguments are created once at definition. The list persists across function calls.

---

78. What is the output?

```
d = {'B': 3}
print(d.get('X', 10))
```

a. 3
b. 10
c. Error
d. None

Correct Answer: 10
Reasoning: The get() method returns the default value (10) because the key 'X' is not in the dictionary.

---

79. What is printed by the last line?

```
def append_to(num, target=[]):
    target.append(num)
    return target

append_to(99)
print(append_to(3))
```

a. [99, 3]
b. [3]
c. Error
d. [99]

Correct Answer: [99, 3]
Reasoning: Default list arguments are created once at definition. The list persists across function calls.

---

80. What is the output?

```
d = {'B': 3}
print(d.get('Z', 10))
```

a. None
b. Error
c. 10
d. 3

Correct Answer: 10
Reasoning: The get() method returns the default value (10) because the key 'Z' is not in the dictionary.

---

81. What is the output of print(type((1, 2)))?

b. Error
c.
d.

Correct Answer:
Reasoning: The value (1, 2) is recognized by Python as .

---

82. What does this evaluate to?

```
print(True and False)
```

a. Error
b. True
c. False
d. None

Correct Answer: False
Reasoning: The 'and' operator requires both operands to be True. True and False is False.

---

83. What is the output?

```
def my_func(**kwargs):
    return len(kwargs)

print(my_func(a=1, b=2, c=3))
```

a. 3
b. Error
c. 0
d. 1

Correct Answer: 3
Reasoning: **kwargs collects keyword arguments into a dictionary. 3 pairs were passed.

---

84. What is the output?

```
nums = [x * 4 for x in range(3)]
print(nums)
```

a. [0, 4, 8]
b. Error
c. [0, 4, 8, 12]
d. [4, 8, 12]

Correct Answer: [0, 4, 8]
Reasoning: range(3) yields 0, 1, 2. Multiplying each by 4 results in [0, 4, 8].

---

85. What is the output?

```
def my_func(**kwargs):
    return len(kwargs)

print(my_func(a=1, b=2, c=3))
```

a. Error
b. 3
c. 1
d. 0

Correct Answer: 3
Reasoning: **kwargs collects keyword arguments into a dictionary. 3 pairs were passed.

---

86. If data = [10, 11, 12, 13, 14, 15, 16], what is the output of print(data[2:5])?

a. [12, 13, 14, 15]
b. Error
c. [12, 13, 14]
d. [13, 14]

Correct Answer: [12, 13, 14]
Reasoning: Slicing [start:end] includes the start index (2) but excludes the end index (5).

---

87. What is the output of print(type([1, 2]))?

a. Error
b.
c.

Correct Answer:
Reasoning: The value [1, 2] is recognized by Python as .

---

88. What is the output?

```
def calc(a, b=5):
    return a + b

print(calc(2))
```

a. 2
b. 7
c. 5
d. 9

Correct Answer: 7
Reasoning: Parameter 'b' defaults to 5. 2 + 5 = 7.

---

89. What is the output?

```
nums = [x * 3 for x in range(3)]
print(nums)
```

a. [0, 3, 6]
b. Error
c. [0, 3, 6, 9]
d. [3, 6, 9]

Correct Answer: [0, 3, 6]
Reasoning: range(3) yields 0, 1, 2. Multiplying each by 3 results in [0, 3, 6].

---

90. What is the output?

```
nums = [x * 3 for x in range(3)]
print(nums)
```

a. Error
b. [0, 3, 6, 9]
c. [0, 3, 6]
d. [3, 6, 9]

Correct Answer: [0, 3, 6]
Reasoning: range(3) yields 0, 1, 2. Multiplying each by 3 results in [0, 3, 6].

---

91. What is the output of print('banana'.find('na'))?

a. 0
b. -1
c. 4
d. 2

Correct Answer: 2
Reasoning: find() returns the lowest index of the substring. 'na' starts at index 2.

---

92. What does this code print?

```
def count_items(*args):
    return len(args)

print(count_items(8, 1, 9, 4, 2, 8))
```

a. 6
b. 7
c. Error
d. 5

Correct Answer: 6
Reasoning: *args collects positional arguments into a tuple. There are 6 arguments passed.

---

93. What is the output?

```
def my_func(**kwargs):
    return len(kwargs)

print(my_func(a=1, b=2, c=3))
```

a. 3
b. 0
c. 1
d. Error

Correct Answer: 3
Reasoning: **kwargs collects keyword arguments into a dictionary. 3 pairs were passed.

---

94. What is the output of print({1, 2} | {2, 3})?

a. {1, 3}
b. {2}
c. {1, 2, 3}
d. Error

Correct Answer: {1, 2, 3}
Reasoning: The | operator performs a set union, combining all unique elements.

---

95. What is the output?

```
nums = [x * 3 for x in range(3)]
print(nums)
```

a. [0, 3, 6]
b. Error
c. [0, 3, 6, 9]
d. [3, 6, 9]

Correct Answer: [0, 3, 6]
Reasoning: range(3) yields 0, 1, 2. Multiplying each by 3 results in [0, 3, 6].

---

96. What is the output?

```
def my_func(**kwargs):
    return len(kwargs)

print(my_func(a=1, b=2, c=3))
```

a. 0
b. 3
c. 1
d. Error

Correct Answer: 3
Reasoning: **kwargs collects keyword arguments into a dictionary. 3 pairs were passed.

---

97. If data = [10, 11, 12, 13, 14, 15, 16], what is the output of print(data[1:3])?

a. Error
b. [11, 12, 13]
c. [11, 12]
d. [12]

Correct Answer: [11, 12]
Reasoning: Slicing [start:end] includes the start index (1) but excludes the end index (3).

---

98. What is the output?

```
d = {'B': 1}
print(d.get('Z', 7))
```

a. None
b. Error
c. 7
d. 1

Correct Answer: 7
Reasoning: The get() method returns the default value (7) because the key 'Z' is not in the dictionary.

---

99. What is printed by the last line?

```
def append_to(num, target=[]):
    target.append(num)
    return target

append_to(99)
print(append_to(2))
```

a. [99]
b. Error
c. [99, 2]
d. [2]

Correct Answer: [99, 2]
Reasoning: Default list arguments are created once at definition. The list persists across function calls.

---

100. What does this evaluate to?

```
print(False and False)
```

a. True
b. None
c. False
d. Error

Correct Answer: False
Reasoning: The 'and' operator requires both operands to be True. False and False is False.

---

101. What is the output of print({1, 2} | {2, 3})?

a. {1, 3}
b. {1, 2, 3}
c. {2}
d. Error

Correct Answer: {1, 2, 3}
Reasoning: The | operator performs a set union, combining all unique elements.

---

102. What is the output of print({1, 2} | {2, 3})?

a. {1, 3}
b. {1, 2, 3}
c. {2}
d. Error

Correct Answer: {1, 2, 3}
Reasoning: The | operator performs a set union, combining all unique elements.

---

103. What is the output?

```
d = {'B': 4}
print(d.get('Y', 10))
```

a. None
b. 10
c. Error
d. 4

Correct Answer: 10
Reasoning: The get() method returns the default value (10) because the key 'Y' is not in the dictionary.

---

104. If data = [10, 11, 12, 13, 14, 15, 16], what is the output of print(data[1:4])?

a. Error
b. [12, 13]
c. [11, 12, 13, 14]
d. [11, 12, 13]

Correct Answer: [11, 12, 13]
Reasoning: Slicing [start:end] includes the start index (1) but excludes the end index (4).

---

105. What is the output of print(type((1, 2)))?

a.
b.
c. Error
d.

Correct Answer:
Reasoning: The value (1, 2) is recognized by Python as .

---

106. What is the output of print(4 + 2 * 3)?

a. 9
b. 18
c. Error
d. 10

Correct Answer: 10
Reasoning: Multiplication has higher precedence than addition, so it evaluates as a + (b * c).

---

107. What does this code print?

```
def count_items(*args):
    return len(args)

print(count_items(3, 1, 5, 7, 6))
```

a. 5
b. 6
c. Error
d. 4

Correct Answer: 5
Reasoning: *args collects positional arguments into a tuple. There are 5 arguments passed.

---

108. What is the output?

```
d = {'B': 4}
print(d.get('Y', 8))
```

a. None
b. Error
c. 4
d. 8

Correct Answer: 8
Reasoning: The get() method returns the default value (8) because the key 'Y' is not in the dictionary.

---

109. What is printed by the last line?

```
def append_to(num, target=[]):
    target.append(num)
    return target

append_to(99)
print(append_to(5))
```

a. [99]
b. Error
c. [5]
d. [99, 5]

Correct Answer: [99, 5]
Reasoning: Default list arguments are created once at definition. The list persists across function calls.

---

110. What is the output of print(type('123'))?

b.
c. Error
d.

Correct Answer:
Reasoning: The value '123' is recognized by Python as .

---

111. What is the output?

```
power = lambda x, y: x ** y
print(power(2, 3))
```

a. 9
b. Error
c. 8
d. 6

Correct Answer: 8
Reasoning: The lambda function returns x raised to the power of y. 2 ** 3 = 8.

---

112. What is the output of print(type('123'))?

b.
c. Error
d.

Correct Answer:
Reasoning: The value '123' is recognized by Python as .

---

113. What is the output?

```
def my_func(**kwargs):
    return len(kwargs)

print(my_func(a=1, b=2, c=3))
```

a. 0
b. Error
c. 3
d. 1

Correct Answer: 3
Reasoning: **kwargs collects keyword arguments into a dictionary. 3 pairs were passed.

---

114. What is the output?

```
def my_func(**kwargs):
    return len(kwargs)

print(my_func(a=1, b=2, c=3))
```

a. 1
b. 3
c. Error
d. 0

Correct Answer: 3
Reasoning: **kwargs collects keyword arguments into a dictionary. 3 pairs were passed.

---

115. What is the output of print(2 + 3 * 4)?

a. 9
b. 14
c. 20
d. Error

Correct Answer: 14
Reasoning: Multiplication has higher precedence than addition, so it evaluates as a + (b * c).

---

116. What is the output?

```
def multiply(a, b=2):
    return a * b

print(multiply(3, 8))
```

a. 6
b. Error
c. 8
d. 24

Correct Answer: 24
Reasoning: The default value 2 is overridden by 8. 3 * 8 = 24.

---

117. What is the output?

```
d = {'C': 3}
print(d.get('Y', 6))
```

a. Error
b. 3
c. 6
d. None

Correct Answer: 6
Reasoning: The get() method returns the default value (6) because the key 'Y' is not in the dictionary.

---

118. What is the output of print({1, 2} | {2, 3})?

a. {1, 3}
b. {2}
c. {1, 2, 3}
d. Error

Correct Answer: {1, 2, 3}
Reasoning: The | operator performs a set union, combining all unique elements.

---

119. What is the output?

```
d = {'C': 2}
print(d.get('Y', 9))
```

a. 9
b. Error
c. None
d. 2

Correct Answer: 9
Reasoning: The get() method returns the default value (9) because the key 'Y' is not in the dictionary.

---

120. What is the output of print(4 + 3 * 3)?

a. 13
b. 21
c. 10
d. Error

Correct Answer: 13
Reasoning: Multiplication has higher precedence than addition, so it evaluates as a + (b * c).

---

121. What is the output?

```
d = {'A': 3}
print(d.get('X', 9))
```

a. 9
b. None
c. Error
d. 3

Correct Answer: 9
Reasoning: The get() method returns the default value (9) because the key 'X' is not in the dictionary.

---

122. What is the output?

```
d = {'A': 3}
print(d.get('Y', 7))
```

a. None
b. 3
c. 7
d. Error

Correct Answer: 7
Reasoning: The get() method returns the default value (7) because the key 'Y' is not in the dictionary.

---

123. What is the output?

```
def my_func(**kwargs):
    return len(kwargs)

print(my_func(a=1, b=2, c=3))
```

a. 3
b. Error
c. 0
d. 1

Correct Answer: 3
Reasoning: **kwargs collects keyword arguments into a dictionary. 3 pairs were passed.

---

124. What is the output of print('banana'.find('na'))?

a. 4
b. 2
c. -1
d. 0

Correct Answer: 2
Reasoning: find() returns the lowest index of the substring. 'na' starts at index 2.

---

125. What is the result of print('b' * 3)?

a. bbb
b. Error
c. 3
d. b

Correct Answer: bbb
Reasoning: Multiplying a string by an integer repeats the string that many times.

---

126. What is the output?

```
nums = [x * 4 for x in range(3)]
print(nums)
```

a. [0, 4, 8]
b. Error
c. [0, 4, 8, 12]
d. [4, 8, 12]

Correct Answer: [0, 4, 8]
Reasoning: range(3) yields 0, 1, 2. Multiplying each by 4 results in [0, 4, 8].

---

127. What is the output of print(type((1, 2)))?

a. <class tuple>
b. Error

Correct Answer:
Reasoning: The value (1, 2) is recognized by Python as .

---

128. If data = [10, 11, 12, 13, 14, 15, 16], what is the output of print(data[0:4])?

a. [10, 11, 12, 13]
b. [11, 12, 13]
c. Error
d. [10, 11, 12, 13, 14]

Correct Answer: [10, 11, 12, 13]
Reasoning: Slicing [start:end] includes the start index (0) but excludes the end index (4).

---

129. What is the output?

```
nums = [x * 4 for x in range(3)]
print(nums)
```

a. [0, 4, 8]
b. Error
c. [4, 8, 12]
d. [0, 4, 8, 12]

Correct Answer: [0, 4, 8]
Reasoning: range(3) yields 0, 1, 2. Multiplying each by 4 results in [0, 4, 8].

---

130. What is the output?

```
d = {'C': 3}
print(d.get('Y', 8))
```

a. 3
b. Error
c. 8
d. None

Correct Answer: 8
Reasoning: The get() method returns the default value (8) because the key 'Y' is not in the dictionary.

---

131. What is printed by the last line?

```
def append_to(num, target=[]):
    target.append(num)
    return target

append_to(99)
print(append_to(2))
```

a. [2]
b. Error
c. [99, 2]
d. [99]

Correct Answer: [99, 2]
Reasoning: Default list arguments are created once at definition. The list persists across function calls.

---

132. What is the output?

```
power = lambda x, y: x ** y
print(power(3, 5))
```

a. 243
b. 15
c. Error
d. 125

Correct Answer: 243
Reasoning: The lambda function returns x raised to the power of y. 3 ** 5 = 243.

---

133. What is the output?

```
def multiply(a, b=4):
    return a * b

print(multiply(3, 7))
```

a. Error
b. 12
c. 21
d. 7

Correct Answer: 21
Reasoning: The default value 4 is overridden by 7. 3 * 7 = 21.

---

134. What is the output?

```
def calc(a, b=4):
    return a + b

print(calc(10))
```

a. 10
b. 16
c. 14
d. 4

Correct Answer: 14
Reasoning: Parameter 'b' defaults to 4. 10 + 4 = 14.

---

135. What is the result of print('c' * 4)?

a. c
b. 4
c. cccc
d. Error

Correct Answer: cccc
Reasoning: Multiplying a string by an integer repeats the string that many times.

---

136. What is the output of print(3 + 3 * 3)?

a. 9
b. 12
c. 18
d. Error

Correct Answer: 12
Reasoning: Multiplication has higher precedence than addition, so it evaluates as a + (b * c).

---

137. What is the output?

```
def calc(a, b=9):
    return a + b

print(calc(6))
```

a. 9
b. 6
c. 17
d. 15

Correct Answer: 15
Reasoning: Parameter 'b' defaults to 9. 6 + 9 = 15.

---

138. What is the output?

```
nums = [x * 2 for x in range(3)]
print(nums)
```

a. [2, 4, 6]
b. [0, 2, 4, 6]
c. Error
d. [0, 2, 4]

Correct Answer: [0, 2, 4]
Reasoning: range(3) yields 0, 1, 2. Multiplying each by 2 results in [0, 2, 4].

---

139. What is the output?

```
def my_func(**kwargs):
    return len(kwargs)

print(my_func(a=1, b=2, c=3))
```

a. 0
b. Error
c. 3
d. 1

Correct Answer: 3
Reasoning: **kwargs collects keyword arguments into a dictionary. 3 pairs were passed.

---

140. What is the output?

```
def multiply(a, b=3):
    return a * b

print(multiply(5, 10))
```

a. 50
b. 10
c. 15
d. Error

Correct Answer: 50
Reasoning: The default value 3 is overridden by 10. 5 * 10 = 50.

---

141. If data = [10, 11, 12, 13, 14, 15, 16], what is the output of print(data[2:4])?

a. [12, 13, 14]
b. [12, 13]
c. Error
d. [13]

Correct Answer: [12, 13]
Reasoning: Slicing [start:end] includes the start index (2) but excludes the end index (4).

---

142. What is the output of print(4 + 2 * 2)?

a. Error
b. 8
c. 41
d. 12

Correct Answer: 8
Reasoning: Multiplication has higher precedence than addition, so it evaluates as a + (b * c).

---

143. What does this code print?

```
def count_items(*args):
    return len(args)

print(count_items(3, 9, 2))
```

a. 4
b. Error
c. 3
d. 2

Correct Answer: 3
Reasoning: *args collects positional arguments into a tuple. There are 3 arguments passed.

---

144. What is the output?

```
nums = [x * 4 for x in range(3)]
print(nums)
```

a. [0, 4, 8, 12]
b. [0, 4, 8]
c. [4, 8, 12]
d. Error

Correct Answer: [0, 4, 8]
Reasoning: range(3) yields 0, 1, 2. Multiplying each by 4 results in [0, 4, 8].

---

145. What is the output?

```
power = lambda x, y: x ** y
print(power(4, 4))
```

a. 16
b. Error
c. 256
d. 45

Correct Answer: 256
Reasoning: The lambda function returns x raised to the power of y. 4 ** 4 = 256.

---

146. What is the output of print('banana'.find('na'))?

a. 4
b. 2
c. -1
d. 0

Correct Answer: 2
Reasoning: find() returns the lowest index of the substring. 'na' starts at index 2.

---

147. What does this evaluate to?

```
print(True and True)
```

a. None
b. True
c. Error
d. False

Correct Answer: True
Reasoning: The 'and' operator requires both operands to be True. True and True is True.

---

148. What is the output?

```
def multiply(a, b=3):
    return a * b

print(multiply(2, 8))
```

a. Error
b. 16
c. 6
d. 8

Correct Answer: 16
Reasoning: The default value 3 is overridden by 8. 2 * 8 = 16.

---

149. What is the output?

```
nums = [x * 4 for x in range(3)]
print(nums)
```

a. [0, 4, 8]
b. [4, 8, 12]
c. Error
d. [0, 4, 8, 12]

Correct Answer: [0, 4, 8]
Reasoning: range(3) yields 0, 1, 2. Multiplying each by 4 results in [0, 4, 8].

---

150. If data = [10, 11, 12, 13, 14, 15, 16], what is the output of print(data[2:5])?

a. [13, 14]
b. [12, 13, 14]
c. [12, 13, 14, 15]
d. Error

Correct Answer: [12, 13, 14]
Reasoning: Slicing [start:end] includes the start index (2) but excludes the end index (5).

---

151. What is the output of print(2 + 3 * 4)?

a. 14
b. 20
c. Error
d. 9

Correct Answer: 14
Reasoning: Multiplication has higher precedence than addition, so it evaluates as a + (b * c).

---

152. What is the output?

```
def calc(a, b=5):
    return a + b

print(calc(3))
```

a. 10
b. 3
c. 5
d. 8

Correct Answer: 8
Reasoning: Parameter 'b' defaults to 5. 3 + 5 = 8.

---

153. What is the output of print(2 + 2 * 3)?

a. 7
b. 8
c. 12
d. Error

Correct Answer: 8
Reasoning: Multiplication has higher precedence than addition, so it evaluates as a + (b * c).

---

154. What is the output of print(3 + 4 * 3)?

a. Error
b. 10
c. 21
d. 15

Correct Answer: 15
Reasoning: Multiplication has higher precedence than addition, so it evaluates as a + (b * c).

---

155. What does this evaluate to?

```
print(False and False)
```

a. None
b. True
c. Error
d. False

Correct Answer: False
Reasoning: The 'and' operator requires both operands to be True. False and False is False.

---

156. What does this evaluate to?

```
print(False and False)
```

a. True
b. False
c. None
d. Error

Correct Answer: False
Reasoning: The 'and' operator requires both operands to be True. False and False is False.

---

157. What is the output?

```
nums = [x * 2 for x in range(3)]
print(nums)
```

a. [0, 2, 4]
b. Error
c. [2, 4, 6]
d. [0, 2, 4, 6]

Correct Answer: [0, 2, 4]
Reasoning: range(3) yields 0, 1, 2. Multiplying each by 2 results in [0, 2, 4].

---

158. What is the output?

```
def multiply(a, b=2):
    return a * b

print(multiply(5, 10))
```

a. 10
b. Error
c. 50
d. 30

Correct Answer: 50
Reasoning: The default value 2 is overridden by 10. 5 * 10 = 50.

---

159. What is the result of print('c' * 6)?

a. c
b. 6
c. cccccc
d. Error

Correct Answer: cccccc
Reasoning: Multiplying a string by an integer repeats the string that many times.

---

160. If data = [10, 11, 12, 13, 14, 15, 16], what is the output of print(data[2:3])?

a. [12]
b. Error
c. [12, 13]
d. []

Correct Answer: [12]
Reasoning: Slicing [start:end] includes the start index (2) but excludes the end index (3).

---

161. What is the output?

```
def multiply(a, b=3):
    return a * b

print(multiply(2, 7))
```

a. 6
b. 14
c. Error
d. 7

Correct Answer: 14
Reasoning: The default value 3 is overridden by 7. 2 * 7 = 14.

---

162. What is the result of print('x' * 3)?

a. 3
b. Error
c. xxx
d. x

Correct Answer: xxx
Reasoning: Multiplying a string by an integer repeats the string that many times.

---

163. What is the output?

```
def multiply(a, b=5):
    return a * b

print(multiply(3, 8))
```

a. 24
b. Error
c. 15
d. 8

Correct Answer: 24
Reasoning: The default value 5 is overridden by 8. 3 * 8 = 24.

---

164. What is the output?

```
nums = [x * 2 for x in range(3)]
print(nums)
```

a. [2, 4, 6]
b. [0, 2, 4]
c. [0, 2, 4, 6]
d. Error

Correct Answer: [0, 2, 4]
Reasoning: range(3) yields 0, 1, 2. Multiplying each by 2 results in [0, 2, 4].

---

165. What does this evaluate to?

```
print(True and True)
```

a. Error
b. None
c. False
d. True

Correct Answer: True
Reasoning: The 'and' operator requires both operands to be True. True and True is True.

---

166. What is the output?

```
d = {'C': 1}
print(d.get('Y', 8))
```

a. 8
b. 1
c. Error
d. None

Correct Answer: 8
Reasoning: The get() method returns the default value (8) because the key 'Y' is not in the dictionary.

---

167. What is the output of print({1, 2} | {2, 3})?

a. {1, 2, 3}
b. {1, 3}
c. Error
d. {2}

Correct Answer: {1, 2, 3}
Reasoning: The | operator performs a set union, combining all unique elements.

---

168. What is the output?

```
d = {'B': 5}
print(d.get('Z', 9))
```

a. None
b. 9
c. 5
d. Error

Correct Answer: 9
Reasoning: The get() method returns the default value (9) because the key 'Z' is not in the dictionary.

---

169. What is printed by the last line?

```
def append_to(num, target=[]):
    target.append(num)
    return target

append_to(99)
print(append_to(2))
```

a. [99]
b. [2]
c. [99, 2]
d. Error

Correct Answer: [99, 2]
Reasoning: Default list arguments are created once at definition. The list persists across function calls.

---

170. What is the output?

```
def calc(a, b=9):
    return a + b

print(calc(7))
```

a. 9
b. 16
c. 7
d. 18

Correct Answer: 16
Reasoning: Parameter 'b' defaults to 9. 7 + 9 = 16.

---

171. What is the output?

```
nums = [x * 2 for x in range(3)]
print(nums)
```

a. [0, 2, 4]
b. Error
c. [0, 2, 4, 6]
d. [2, 4, 6]

Correct Answer: [0, 2, 4]
Reasoning: range(3) yields 0, 1, 2. Multiplying each by 2 results in [0, 2, 4].

---

172. What is the output of print({1, 2} | {2, 3})?

a. {1, 3}
b. Error
c. {2}
d. {1, 2, 3}

Correct Answer: {1, 2, 3}
Reasoning: The | operator performs a set union, combining all unique elements.

---

173. What is the result of print('x' * 3)?

a. 3
b. xxx
c. Error
d. x

Correct Answer: xxx
Reasoning: Multiplying a string by an integer repeats the string that many times.

---

174. What is the output?

```
def multiply(a, b=4):
    return a * b

print(multiply(5, 8))
```

a. 20
b. 40
c. 8
d. Error

Correct Answer: 40
Reasoning: The default value 4 is overridden by 8. 5 * 8 = 40.

---

175. What is the output of print({1, 2} | {2, 3})?

a. {1, 2, 3}
b. {2}
c. Error
d. {1, 3}

Correct Answer: {1, 2, 3}
Reasoning: The | operator performs a set union, combining all unique elements.

---

176. What is the output?

```
d = {'C': 4}
print(d.get('Z', 10))
```

a. 4
b. Error
c. None
d. 10

Correct Answer: 10
Reasoning: The get() method returns the default value (10) because the key 'Z' is not in the dictionary.

---

177. What is the output?

```
power = lambda x, y: x ** y
print(power(4, 3))
```

a. 81
b. Error
c. 12
d. 64

Correct Answer: 64
Reasoning: The lambda function returns x raised to the power of y. 4 ** 3 = 64.

---

178. What is the output?

```
def my_func(**kwargs):
    return len(kwargs)

print(my_func(a=1, b=2, c=3))
```

a. 1
b. Error
c. 0
d. 3

Correct Answer: 3
Reasoning: **kwargs collects keyword arguments into a dictionary. 3 pairs were passed.

---

179. What is printed by the last line?

```
def append_to(num, target=[]):
    target.append(num)
    return target

append_to(99)
print(append_to(3))
```

a. [99]
b. [3]
c. Error
d. [99, 3]

Correct Answer: [99, 3]
Reasoning: Default list arguments are created once at definition. The list persists across function calls.

---

180. What is the output?

```
def my_func(**kwargs):
    return len(kwargs)

print(my_func(a=1, b=2, c=3))
```

a. Error
b. 1
c. 0
d. 3

Correct Answer: 3
Reasoning: **kwargs collects keyword arguments into a dictionary. 3 pairs were passed.

---

181. What is the output?

```
d = {'B': 2}
print(d.get('X', 10))
```

a. 2
b. 10
c. None
d. Error

Correct Answer: 10
Reasoning: The get() method returns the default value (10) because the key 'X' is not in the dictionary.

---

182. What does this code print?

```
def count_items(*args):
    return len(args)

print(count_items(2, 7, 4))
```

a. 2
b. 3
c. 4
d. Error

Correct Answer: 3
Reasoning: *args collects positional arguments into a tuple. There are 3 arguments passed.

---

183. What is the output?

```
power = lambda x, y: x ** y
print(power(5, 4))
```

a. 1024
b. 625
c. 20
d. Error

Correct Answer: 625
Reasoning: The lambda function returns x raised to the power of y. 5 ** 4 = 625.

---

184. What is the output of print('banana'.find('na'))?

a. 0
b. 2
c. 4
d. -1

Correct Answer: 2
Reasoning: find() returns the lowest index of the substring. 'na' starts at index 2.

---

185. What is the output?

```
def multiply(a, b=4):
    return a * b

print(multiply(5, 10))
```

a. 50
b. 20
c. Error
d. 10

Correct Answer: 50
Reasoning: The default value 4 is overridden by 10. 5 * 10 = 50.

---

186. What is printed by the last line?

```
def append_to(num, target=[]):
    target.append(num)
    return target

append_to(99)
print(append_to(3))
```

a. [99]
b. [99, 3]
c. [3]
d. Error

Correct Answer: [99, 3]
Reasoning: Default list arguments are created once at definition. The list persists across function calls.

---

187. What does this evaluate to?

```
print(False and True)
```

a. False
b. None
c. Error
d. True

Correct Answer: False
Reasoning: The 'and' operator requires both operands to be True. False and True is False.

---

188. What is the output of print(3 + 4 * 3)?

a. Error
b. 21
c. 10
d. 15

Correct Answer: 15
Reasoning: Multiplication has higher precedence than addition, so it evaluates as a + (b * c).

---

189. If data = [10, 11, 12, 13, 14, 15, 16], what is the output of print(data[0:5])?

a. Error
b. [10, 11, 12, 13, 14, 15]
c. [11, 12, 13, 14]
d. [10, 11, 12, 13, 14]

Correct Answer: [10, 11, 12, 13, 14]
Reasoning: Slicing [start:end] includes the start index (0) but excludes the end index (5).

---

190. What is the output of print({1, 2} | {2, 3})?

a. Error
b. {2}
c. {1, 2, 3}
d. {1, 3}

Correct Answer: {1, 2, 3}
Reasoning: The | operator performs a set union, combining all unique elements.

---

191. What is the output?

```
power = lambda x, y: x ** y
print(power(3, 4))
```

a. Error
b. 81
c. 64
d. 12

Correct Answer: 81
Reasoning: The lambda function returns x raised to the power of y. 3 ** 4 = 81.

---

192. What is the output of print('banana'.find('na'))?

a. -1
b. 0
c. 4
d. 2

Correct Answer: 2
Reasoning: find() returns the lowest index of the substring. 'na' starts at index 2.

---

193. What is the output?

```
def calc(a, b=4):
    return a + b

print(calc(7))
```

a. 13
b. 4
c. 7
d. 11

Correct Answer: 11
Reasoning: Parameter 'b' defaults to 4. 7 + 4 = 11.

---

194. What is printed by the last line?

```
def append_to(num, target=[]):
    target.append(num)
    return target

append_to(99)
print(append_to(4))
```

a. Error
b. [99]
c. [99, 4]
d. [4]

Correct Answer: [99, 4]
Reasoning: Default list arguments are created once at definition. The list persists across function calls.

---

195. What is the output?

```
def my_func(**kwargs):
    return len(kwargs)

print(my_func(a=1, b=2, c=3))
```

a. 3
b. 0
c. Error
d. 1

Correct Answer: 3
Reasoning: **kwargs collects keyword arguments into a dictionary. 3 pairs were passed.

---

196. What is the output of print({1, 2} | {2, 3})?

a. {1, 2, 3}
b. {1, 3}
c. Error
d. {2}

Correct Answer: {1, 2, 3}
Reasoning: The | operator performs a set union, combining all unique elements.

---

197. What is the output?

```
def my_func(**kwargs):
    return len(kwargs)

print(my_func(a=1, b=2, c=3))
```

a. 3
b. 0
c. Error
d. 1

Correct Answer: 3
Reasoning: **kwargs collects keyword arguments into a dictionary. 3 pairs were passed.

---

198. What is the output?

```
def my_func(**kwargs):
    return len(kwargs)

print(my_func(a=1, b=2, c=3))
```

a. 1
b. Error
c. 0
d. 3

Correct Answer: 3
Reasoning: **kwargs collects keyword arguments into a dictionary. 3 pairs were passed.

---

199. What does this code print?

```
def count_items(*args):
    return len(args)

print(count_items(7, 9, 6))
```

a. 4
b. Error
c. 2
d. 3

Correct Answer: 3
Reasoning: *args collects positional arguments into a tuple. There are 3 arguments passed.

---

200. If data = [10, 11, 12, 13, 14, 15, 16], what is the output of print(data[2:3])?

a. [12]
b. []
c. [12, 13]
d. Error

Correct Answer: [12]
Reasoning: Slicing [start:end] includes the start index (2) but excludes the end index (3).

---

201. What is the output of print(3 + 4 * 3)?

a. 21
b. Error
c. 10
d. 15

Correct Answer: 15
Reasoning: Multiplication has higher precedence than addition, so it evaluates as a + (b * c).

---

202. What is the output?

```
nums = [x * 2 for x in range(3)]
print(nums)
```

a. [2, 4, 6]
b. [0, 2, 4]
c. Error
d. [0, 2, 4, 6]

Correct Answer: [0, 2, 4]
Reasoning: range(3) yields 0, 1, 2. Multiplying each by 2 results in [0, 2, 4].

---

203. What is the output of print(type('123'))?

a.
b.
c. Error

Correct Answer:
Reasoning: The value '123' is recognized by Python as .

---

204. What is the output of print(type((1, 2)))?

a.
b.
c. Error
d.

Correct Answer:
Reasoning: The value (1, 2) is recognized by Python as .

---

205. What is the output?

```
def calc(a, b=9):
    return a + b

print(calc(6))
```

a. 6
b. 9
c. 17
d. 15

Correct Answer: 15
Reasoning: Parameter 'b' defaults to 9. 6 + 9 = 15.

---

206. What is the output of print('banana'.find('na'))?

a. 2
b. 0
c. -1
d. 4

Correct Answer: 2
Reasoning: find() returns the lowest index of the substring. 'na' starts at index 2.

---

207. What is the output?

```
d = {'C': 5}
print(d.get('Z', 9))
```

a. None
b. 9
c. Error
d. 5

Correct Answer: 9
Reasoning: The get() method returns the default value (9) because the key 'Z' is not in the dictionary.

---

208. What is the output of print({1, 2} | {2, 3})?

a. {1, 2, 3}
b. {2}
c. {1, 3}
d. Error

Correct Answer: {1, 2, 3}
Reasoning: The | operator performs a set union, combining all unique elements.

---

209. If data = [10, 11, 12, 13, 14, 15, 16], what is the output of print(data[0:5])?

a. Error
b. [10, 11, 12, 13, 14, 15]
c. [10, 11, 12, 13, 14]
d. [11, 12, 13, 14]

Correct Answer: [10, 11, 12, 13, 14]
Reasoning: Slicing [start:end] includes the start index (0) but excludes the end index (5).

---

210. What is the output?

```
nums = [x * 2 for x in range(3)]
print(nums)
```

a. Error
b. [2, 4, 6]
c. [0, 2, 4]
d. [0, 2, 4, 6]

Correct Answer: [0, 2, 4]
Reasoning: range(3) yields 0, 1, 2. Multiplying each by 2 results in [0, 2, 4].

---

211. What is the result of print('c' * 3)?

a. 3
b. ccc
c. c
d. Error

Correct Answer: ccc
Reasoning: Multiplying a string by an integer repeats the string that many times.

---

212. What is the output?

```
def multiply(a, b=2):
    return a * b

print(multiply(5, 7))
```

a. 7
b. 10
c. Error
d. 35

Correct Answer: 35
Reasoning: The default value 2 is overridden by 7. 5 * 7 = 35.

---

213. What is the output?

```
power = lambda x, y: x ** y
print(power(3, 3))
```

a. 5
b. 27
c. Error
d. 9

Correct Answer: 27
Reasoning: The lambda function returns x raised to the power of y. 3 ** 3 = 27.

---

214. What is the output of print({1, 2} | {2, 3})?

a. {1, 3}
b. {1, 2, 3}
c. Error
d. {2}

Correct Answer: {1, 2, 3}
Reasoning: The | operator performs a set union, combining all unique elements.

---

215. What is the output?

```
def my_func(**kwargs):
    return len(kwargs)

print(my_func(a=1, b=2, c=3))
```

a. 1
b. Error
c. 0
d. 3

Correct Answer: 3
Reasoning: **kwargs collects keyword arguments into a dictionary. 3 pairs were passed.

---

216. What does this code print?

```
def count_items(*args):
    return len(args)

print(count_items(5, 3, 2, 1, 3, 1))
```

a. Error
b. 5
c. 6
d. 7

Correct Answer: 6
Reasoning: *args collects positional arguments into a tuple. There are 6 arguments passed.

---

217. What is the output?

```
def calc(a, b=6):
    return a + b

print(calc(4))
```

a. 6
b. 4
c. 12
d. 10

Correct Answer: 10
Reasoning: Parameter 'b' defaults to 6. 4 + 6 = 10.

---

218. What is the result of print('a' * 3)?

a. aaa
b. 3
c. a
d. Error

Correct Answer: aaa
Reasoning: Multiplying a string by an integer repeats the string that many times.

---

219. What is the result of print('y' * 3)?

a. y
b. yyy
c. 3
d. Error

Correct Answer: yyy
Reasoning: Multiplying a string by an integer repeats the string that many times.

---

220. What is the output of print(2 + 4 * 4)?

a. Error
b. 18
c. 24
d. 10

Correct Answer: 18
Reasoning: Multiplication has higher precedence than addition, so it evaluates as a + (b * c).

---

221. What is the output?

```
def calc(a, b=8):
    return a + b

print(calc(9))
```

a. 9
b. 17
c. 8
d. 19

Correct Answer: 17
Reasoning: Parameter 'b' defaults to 8. 9 + 8 = 17.

---

222. What is the output?

```
nums = [x * 4 for x in range(3)]
print(nums)
```

a. [4, 8, 12]
b. [0, 4, 8, 12]
c. [0, 4, 8]
d. Error

Correct Answer: [0, 4, 8]
Reasoning: range(3) yields 0, 1, 2. Multiplying each by 4 results in [0, 4, 8].

---

223. What is the output?

```
d = {'C': 1}
print(d.get('X', 7))
```

a. 7
b. 1
c. Error
d. None

Correct Answer: 7
Reasoning: The get() method returns the default value (7) because the key 'X' is not in the dictionary.

---

224. What does this evaluate to?

```
print(False and False)
```

a. True
b. None
c. False
d. Error

Correct Answer: False
Reasoning: The 'and' operator requires both operands to be True. False and False is False.

---

225. What is the output of print({1, 2} | {2, 3})?

a. {1, 2, 3}
b. {1, 3}
c. Error
d. {2}

Correct Answer: {1, 2, 3}
Reasoning: The | operator performs a set union, combining all unique elements.

---

226. What is the output of print('banana'.find('na'))?

a. 2
b. 0
c. -1
d. 4

Correct Answer: 2
Reasoning: find() returns the lowest index of the substring. 'na' starts at index 2.

---

227. What is the output of print({1, 2} | {2, 3})?

a. {1, 2, 3}
b. {1, 3}
c. {2}
d. Error

Correct Answer: {1, 2, 3}
Reasoning: The | operator performs a set union, combining all unique elements.

---

228. What is printed by the last line?

```
def append_to(num, target=[]):
    target.append(num)
    return target

append_to(99)
print(append_to(1))
```

a. [99, 1]
b. Error
c. [1]
d. [99]

Correct Answer: [99, 1]
Reasoning: Default list arguments are created once at definition. The list persists across function calls.

---

229. What is the output of print(type([1, 2]))?

a. Error
b.
c.

Correct Answer:
Reasoning: The value [1, 2] is recognized by Python as .

---

230. What is the output?

```
def multiply(a, b=2):
    return a * b

print(multiply(2, 6))
```

a. 6
b. 4
c. 12
d. Error

Correct Answer: 12
Reasoning: The default value 2 is overridden by 6. 2 * 6 = 12.

---

231. What is the output?

```
def my_func(**kwargs):
    return len(kwargs)

print(my_func(a=1, b=2, c=3))
```

a. Error
b. 3
c. 1
d. 0

Correct Answer: 3
Reasoning: **kwargs collects keyword arguments into a dictionary. 3 pairs were passed.

---

232. What is the output of print(4 + 4 * 3)?

a. Error
b. 11
c. 24
d. 16

Correct Answer: 16
Reasoning: Multiplication has higher precedence than addition, so it evaluates as a + (b * c).

---

233. What is the output of print({1, 2} | {2, 3})?

a. {2}
b. {1, 2, 3}
c. {1, 3}
d. Error

Correct Answer: {1, 2, 3}
Reasoning: The | operator performs a set union, combining all unique elements.

---

234. If data = [10, 11, 12, 13, 14, 15, 16], what is the output of print(data[2:4])?

a. [12, 13, 14]
b. Error
c. [12, 13]
d. [13]

Correct Answer: [12, 13]
Reasoning: Slicing [start:end] includes the start index (2) but excludes the end index (4).

---

235. What does this code print?

```
def count_items(*args):
    return len(args)

print(count_items(4, 2, 6, 9, 6))
```

a. 6
b. Error
c. 4
d. 5

Correct Answer: 5
Reasoning: *args collects positional arguments into a tuple. There are 5 arguments passed.

---

236. What is the output?

```
def my_func(**kwargs):
    return len(kwargs)

print(my_func(a=1, b=2, c=3))
```

a. 0
b. 3
c. 1
d. Error

Correct Answer: 3
Reasoning: **kwargs collects keyword arguments into a dictionary. 3 pairs were passed.

---

237. What is the output of print(4 + 4 * 3)?

a. 11
b. 24
c. Error
d. 16

Correct Answer: 16
Reasoning: Multiplication has higher precedence than addition, so it evaluates as a + (b * c).

---

238. What is printed by the last line?

```
def append_to(num, target=[]):
    target.append(num)
    return target

append_to(99)
print(append_to(3))
```

a. Error
b. [99, 3]
c. [3]
d. [99]

Correct Answer: [99, 3]
Reasoning: Default list arguments are created once at definition. The list persists across function calls.

---

239. What does this code print?

```
def count_items(*args):
    return len(args)

print(count_items(7, 2, 3))
```

a. 2
b. 4
c. Error
d. 3

Correct Answer: 3
Reasoning: *args collects positional arguments into a tuple. There are 3 arguments passed.

---

240. What does this evaluate to?

```
print(False and False)
```

a. Error
b. True
c. False
d. None

Correct Answer: False
Reasoning: The 'and' operator requires both operands to be True. False and False is False.

---

241. If data = [10, 11, 12, 13, 14, 15, 16], what is the output of print(data[2:4])?

a. [12, 13, 14]
b. [13]
c. Error
d. [12, 13]

Correct Answer: [12, 13]
Reasoning: Slicing [start:end] includes the start index (2) but excludes the end index (4).

---

242. What does this code print?

```
def count_items(*args):
    return len(args)

print(count_items(1, 2, 5, 8, 7, 8))
```

a. 6
b. 7
c. 5
d. Error

Correct Answer: 6
Reasoning: *args collects positional arguments into a tuple. There are 6 arguments passed.

---

243. What is the output?

```
power = lambda x, y: x ** y
print(power(2, 4))
```

a. 28
b. Error
c. 8
d. 16

Correct Answer: 16
Reasoning: The lambda function returns x raised to the power of y. 2 ** 4 = 16.

---

244. What is the result of print('c' * 3)?

a. 3
b. ccc
c. Error
d. c

Correct Answer: ccc
Reasoning: Multiplying a string by an integer repeats the string that many times.

---

245. What is the output?

```
def calc(a, b=6):
    return a + b

print(calc(5))
```

a. 6
b. 13
c. 11
d. 5

Correct Answer: 11
Reasoning: Parameter 'b' defaults to 6. 5 + 6 = 11.

---

246. If data = [10, 11, 12, 13, 14, 15, 16], what is the output of print(data[1:3])?

a. Error
b. [12]
c. [11, 12, 13]
d. [11, 12]

Correct Answer: [11, 12]
Reasoning: Slicing [start:end] includes the start index (1) but excludes the end index (3).

---

247. What does this code print?

```
def count_items(*args):
    return len(args)

print(count_items(7, 9, 4))
```

a. Error
b. 2
c. 3
d. 4

Correct Answer: 3
Reasoning: *args collects positional arguments into a tuple. There are 3 arguments passed.

---

248. What is the output of print({1, 2} | {2, 3})?

a. {1, 3}
b. Error
c. {2}
d. {1, 2, 3}

Correct Answer: {1, 2, 3}
Reasoning: The | operator performs a set union, combining all unique elements.

---

249. What is the output?

```
nums = [x * 2 for x in range(3)]
print(nums)
```

a. [0, 2, 4]
b. [2, 4, 6]
c. Error
d. [0, 2, 4, 6]

Correct Answer: [0, 2, 4]
Reasoning: range(3) yields 0, 1, 2. Multiplying each by 2 results in [0, 2, 4].

---

250. What is the output of print(3 + 4 * 4)?

a. 11
b. 19
c. 28
d. Error

Correct Answer: 19
Reasoning: Multiplication has higher precedence than addition, so it evaluates as a + (b * c).

---

251. What does this evaluate to?

```
print(True and False)
```

a. False
b. Error
c. None
d. True

Correct Answer: False
Reasoning: The 'and' operator requires both operands to be True. True and False is False.

---

252. What is the output of print({1, 2} | {2, 3})?

a. Error
b. {2}
c. {1, 2, 3}
d. {1, 3}

Correct Answer: {1, 2, 3}
Reasoning: The | operator performs a set union, combining all unique elements.

---

253. What is the output of print({1, 2} | {2, 3})?

a. {1, 3}
b. {2}
c. Error
d. {1, 2, 3}

Correct Answer: {1, 2, 3}
Reasoning: The | operator performs a set union, combining all unique elements.

254. What is the output of print(3 + 4 * 2)?

a. 11
b. 9
c. 14
d. Error

Correct Answer: 11
Reasoning: Multiplication has higher precedence than addition, so it evaluates as a + (b * c).

255. What is the output?

```
power = lambda x, y: x ** y
print(power(4, 3))
```

a. 12
b. Error
c. 81
d. 64

Correct Answer: 64
Reasoning: The lambda function returns x raised to the power of y. 4 ** 3 = 64.

256. What is the result of print('y' * 4)?

a. Error
b. yyyy
c. y
d. 4

Correct Answer: yyyy
Reasoning: Multiplying a string by an integer repeats the string that many times.

257. What is the output?

```
power = lambda x, y: x ** y
print(power(2, 3))
```

a. 6
b. Error
c. 8
d. 9

Correct Answer: 8
Reasoning: The lambda function returns x raised to the power of y. 2 ** 3 = 8.

---

258. What is the output of print(2 + 4 * 2)?

a. 10
b. Error
c. 8
d. 12

Correct Answer: 10
Reasoning: Multiplication has higher precedence than addition, so it evaluates as a + (b * c).

---

259. What is the output?

```
nums = [x * 4 for x in range(3)]
print(nums)
```

a. [0, 4, 8]
b. [4, 8, 12]
c. [0, 4, 8, 12]
d. Error

Correct Answer: [0, 4, 8]
Reasoning: range(3) yields 0, 1, 2. Multiplying each by 4 results in [0, 4, 8].

---

260. What is the output of print({1, 2} | {2, 3})?

a. Error
b. {2}
c. {1, 2, 3}
d. {1, 3}

Correct Answer: {1, 2, 3}
Reasoning: The | operator performs a set union, combining all unique elements.

---

261. What is the output?

```
def multiply(a, b=5):
    return a * b

print(multiply(5, 8))
```

a. 25
b. 40
c. Error
d. 8

Correct Answer: 40
Reasoning: The default value 5 is overridden by 8. 5 * 8 = 40.

---

262. What is the output?

```
d = {'C': 3}
print(d.get('X', 9))
```

a. 3
b. None
c. 9
d. Error

Correct Answer: 9
Reasoning: The get() method returns the default value (9) because the key 'X' is not in the dictionary.

---

263. What is the output?

```
d = {'B': 1}
print(d.get('X', 10))
```

a. None
b. 10
c. 1
d. Error

Correct Answer: 10
Reasoning: The get() method returns the default value (10) because the key 'X' is not in the dictionary.

---

264. What is the output?

```
nums = [x * 4 for x in range(3)]
print(nums)
```

a. [0, 4, 8]
b. [0, 4, 8, 12]
c. Error
d. [4, 8, 12]

Correct Answer: [0, 4, 8]
Reasoning: range(3) yields 0, 1, 2. Multiplying each by 4 results in [0, 4, 8].

---

265. What is the result of print('b' * 3)?

a. 3
b. b
c. bbb
d. Error

Correct Answer: bbb
Reasoning: Multiplying a string by an integer repeats the string that many times.

---

266. What is printed by the last line?

```
def append_to(num, target=[]):
    target.append(num)
    return target

append_to(99)
print(append_to(1))
```

a. Error
b. [99]
c. [99, 1]
d. [1]

Correct Answer: [99, 1]
Reasoning: Default list arguments are created once at definition. The list persists across function calls.

---

267. What is printed by the last line?

```
def append_to(num, target=[]):
    target.append(num)
    return target

append_to(99)
print(append_to(4))
```

a. [4]
b. [99, 4]
c. [99]
d. Error

Correct Answer: [99, 4]
Reasoning: Default list arguments are created once at definition. The list persists across function calls.

---

268. What is the output?

```
d = {'B': 4}
print(d.get('Y', 9))
```

a. 4
b. 9
c. Error
d. None

Correct Answer: 9
Reasoning: The get() method returns the default value (9) because the key 'Y' is not in the dictionary.

---

269. What is the output?

```
def my_func(**kwargs):
    return len(kwargs)

print(my_func(a=1, b=2, c=3))
```

a. Error
b. 0
c. 3
d. 1

Correct Answer: 3
Reasoning: **kwargs collects keyword arguments into a dictionary. 3 pairs were passed.

---

270. What is the output?

```
def multiply(a, b=4):
    return a * b

print(multiply(5, 8))
```

a. 8
b. 40
c. Error
d. 20

Correct Answer: 40
Reasoning: The default value 4 is overridden by 8. 5 * 8 = 40.

---

271. What is the output of print(4 + 2 * 3)?

a. 10
b. 18
c. Error
d. 9

Correct Answer: 10
Reasoning: Multiplication has higher precedence than addition, so it evaluates as a + (b * c).

---

272. What does this code print?

```
def count_items(*args):
    return len(args)

print(count_items(2, 5, 9, 7, 6, 3))
```

a. Error
b. 5
c. 7
d. 6

Correct Answer: 6
Reasoning: *args collects positional arguments into a tuple. There are 6 arguments passed.

---

273. What is the result of print('c' * 3)?

a. Error
b. ccc
c. c
d. 3

Correct Answer: ccc
Reasoning: Multiplying a string by an integer repeats the string that many times.

---

274. What is the output?

```
power = lambda x, y: x ** y
print(power(4, 5))
```

a. Error
b. 625
c. 1024
d. 20

Correct Answer: 1024
Reasoning: The lambda function returns x raised to the power of y. 4 ** 5 = 1024.

---

275. What does this code print?

```
def count_items(*args):
    return len(args)

print(count_items(5, 2))
```

a. Error
b. 1
c. 3
d. 2

Correct Answer: 2
Reasoning: *args collects positional arguments into a tuple. There are 2 arguments passed.

---

276. What is the output of print('banana'.find('na'))?

a. 2
b. -1
c. 0
d. 4

Correct Answer: 2
Reasoning: find() returns the lowest index of the substring. 'na' starts at index 2.

---

277. What does this code print?

```
def count_items(*args):
    return len(args)

print(count_items(3, 3, 8))
```

a. Error
b. 4
c. 2
d. 3

Correct Answer: 3
Reasoning: *args collects positional arguments into a tuple. There are 3 arguments passed.

---

278. What is the output of print(type(45.0))?

a.
b.
c. Error

Correct Answer:
Reasoning: The value 45.0 is recognized by Python as .

---

279. What is the output?

```
def my_func(**kwargs):
    return len(kwargs)

print(my_func(a=1, b=2, c=3))
```

a. 0
b. 1
c. 3
d. Error

Correct Answer: 3
Reasoning: **kwargs collects keyword arguments into a dictionary. 3 pairs were passed.

---

280. If data = [10, 11, 12, 13, 14, 15, 16], what is the output of print(data[1:3])?

a. [11, 12]
b. [11, 12, 13]
c. Error
d. [12]

Correct Answer: [11, 12]
Reasoning: Slicing [start:end] includes the start index (1) but excludes the end index (3).

---

281. What is the output?

```
nums = [x * 3 for x in range(3)]
print(nums)
```

a. [0, 3, 6]
b. [3, 6, 9]
c. [0, 3, 6, 9]
d. Error

Correct Answer: [0, 3, 6]
Reasoning: range(3) yields 0, 1, 2. Multiplying each by 3 results in [0, 3, 6].

---

282. What does this code print?

```
def count_items(*args):
    return len(args)

print(count_items(3, 5, 1))
```

a. 4
b. 3
c. 2
d. Error

Correct Answer: 3
Reasoning: *args collects positional arguments into a tuple. There are 3 arguments passed.

---

283. What is printed by the last line?

```
def append_to(num, target=[]):
    target.append(num)
    return target

append_to(99)
print(append_to(5))
```

a. [99]
b. Error
c. [5]
d. [99, 5]

Correct Answer: [99, 5]
Reasoning: Default list arguments are created once at definition. The list persists across function calls.

---

284. What is the output?

```
def my_func(**kwargs):
    return len(kwargs)

print(my_func(a=1, b=2, c=3))
```

a. 3
b. 1
c. Error
d. 0

Correct Answer: 3
Reasoning: **kwargs collects keyword arguments into a dictionary. 3 pairs were passed.

---

285. What is the output of print('banana'.find('na'))?

a. 0
b. 2
c. -1
d. 4

Correct Answer: 2
Reasoning: find() returns the lowest index of the substring. 'na' starts at index 2.

---

286. What is the output of print('banana'.find('na'))?

a. 2
b. 0
c. -1
d. 4

Correct Answer: 2
Reasoning: find() returns the lowest index of the substring. 'na' starts at index 2.

---

287. What is the output of print('banana'.find('na'))?

a. 0
b. -1
c. 4
d. 2

Correct Answer: 2
Reasoning: find() returns the lowest index of the substring. 'na' starts at index 2.

---

288. What does this evaluate to?

```
print(False and True)
```

a. None
b. True
c. False
d. Error

Correct Answer: False
Reasoning: The 'and' operator requires both operands to be True. False and True is False.

---

289. What is the output?

```
power = lambda x, y: x ** y
print(power(2, 4))
```

a. 8
b. 7
c. 16
d. Error

Correct Answer: 16
Reasoning: The lambda function returns x raised to the power of y. 2 ** 4 = 16.

---

290. What is the output?

```
def calc(a, b=9):
    return a + b

print(calc(6))
```

a. 15
b. 17
c. 9
d. 6

Correct Answer: 15
Reasoning: Parameter 'b' defaults to 9. 6 + 9 = 15.

---

291. What is the output?

```
nums = [x * 4 for x in range(3)]
print(nums)
```

a. [4, 8, 12]
b. [0, 4, 8, 12]
c. Error
d. [0, 4, 8]

Correct Answer: [0, 4, 8]
Reasoning: range(3) yields 0, 1, 2. Multiplying each by 4 results in [0, 4, 8].

---

292. If data = [10, 11, 12, 13, 14, 15, 16], what is the output of print(data[0:3])?

a. [11, 12]
b. [10, 11, 12, 13]
c. [10, 11, 12]
d. Error

Correct Answer: [10, 11, 12]
Reasoning: Slicing [start:end] includes the start index (0) but excludes the end index (3).

---

293. What is the output?

```
power = lambda x, y: x ** y
print(power(4, 2))
```

a. 16
b. Error
c. 4
d. 8

Correct Answer: 16
Reasoning: The lambda function returns x raised to the power of y. 4 ** 2 = 16.

---

294. What is the output?

```
def multiply(a, b=5):
    return a * b

print(multiply(2, 10))
```

a. 10
b. 35
c. Error
d. 20

Correct Answer: 20
Reasoning: The default value 5 is overridden by 10. 2 * 10 = 20.

---

295. What is printed by the last line?

```
def append_to(num, target=[]):
    target.append(num)
    return target

append_to(99)
print(append_to(2))
```

a. Error
b. [99, 2]
c. [2]
d. [99]

Correct Answer: [99, 2]
Reasoning: Default list arguments are created once at definition. The list persists across function calls.

---

296. What is the output of print(3 + 2 * 4)?

a. 20
b. 9
c. 11
d. Error

Correct Answer: 11
Reasoning: Multiplication has higher precedence than addition, so it evaluates as a + (b * c).

---

297. What is the result of print('c' * 3)?

a. Error
b. 3
c. c
d. ccc

Correct Answer: ccc
Reasoning: Multiplying a string by an integer repeats the string that many times.

---

298. What is the output?

```
def calc(a, b=7):
    return a + b

print(calc(8))
```

a. 8
b. 15
c. 17
d. 7

Correct Answer: 15
Reasoning: Parameter 'b' defaults to 7. 8 + 7 = 15.

---

299. What is the output?

```
d = {'A': 2}
print(d.get('Y', 6))
```

a. 2
b. Error
c. 6
d. None

Correct Answer: 6
Reasoning: The get() method returns the default value (6) because the key 'Y' is not in the dictionary.

---

300. What is the output?

```
d = {'C': 2}
print(d.get('Y', 9))
```

a. 2
b. 9
c. None
d. Error

Correct Answer: 9
Reasoning: The get() method returns the default value (9) because the key 'Y' is not in the dictionary.